

close eye on market microstructure. Decimalization has eroded scalpers' profits so the imperative on keeping things simple and getting the market data and execution architecture in place is ever increased. Scalping is the least quantitatively involved strategy which we are covering. To ensure it works well one needs to have a good quantitative model of market direction on which there is a plethora of literature—see for instance the application of neural networks (Resta, 2006), traditional time-series analysis techniques (Dacorogna et al., 2001), and sequential Monte-Carlo methods (Hanif and Smith, 2012b; Javaheri, 2005).

Complementary to the above, there are two further types of scalping (Graifer, 2005). Volume scalping is usually achieved through purchasing a large number of shares for a small gain and betting on an extremely small price improvement. This type of scalping works only on highly liquid stocks. The third type of scalping is closest to traditional trading. The trader enters into a trade and closes out as soon as the first exit condition, usually when the 1:1 risk/reward is met.

Scalping is affected by liquidity, volatility, time horizons and risk management. As mentioned above, some traders favor liquid markets, whilst others favor illiquid markets. Volatility is a specific concern of novice scalpers; however, professional scalpers are indifferent to periods of volatility as a thoroughly calibrated directional model should enable the scalper to benefit from both upswings and downturns. Scalpers typically operate in timeframes beyond common technical analysis, and hence require bespoke tools to enable them to capitalize on microstructural changes. Finally, it is imperative that robust risk management protocols are in place to avoid any accumulations of losses and losing streaks (Graifer, 2005).

It would be prudent to note, financial advisers engaged in scalping are frequently found guilty of market manipulation. Scalping by traders is akin to front-running by advisers (buy a security and subsequently advise a purchase) and has been ruled illegal by the US Supreme Court with the SEC taking a blunter view and banning any scalping where a trust relationship exists between the trader and recommendee (Court, 1963; Commission, 2012).

Market-Neutral Arbitrage

Market-neutral arbitrage is a class of strategies based on fundamental theories of finance. An active strategy, the trader seeks to capitalize on stock selection as a long-short investment strategy (Fabozzi et al., 2008). In contrast to traditional investing, the trader here is aiming to use insights about both stocks which are expected to outperform and those